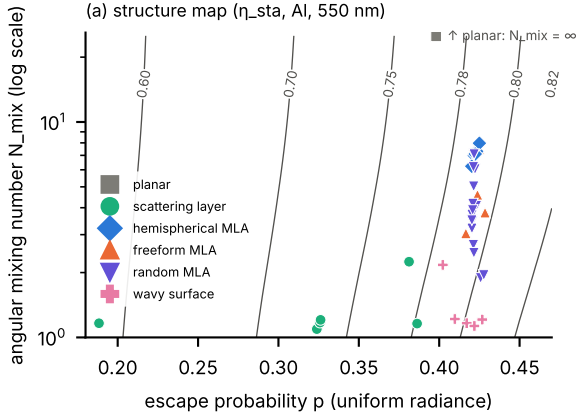
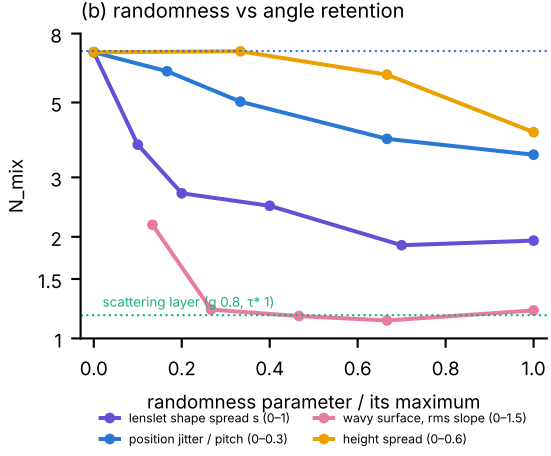
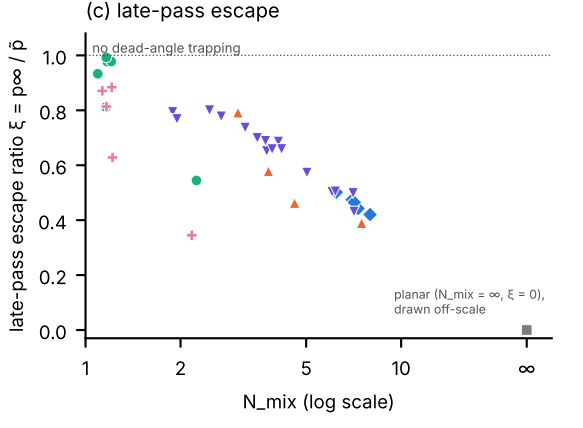
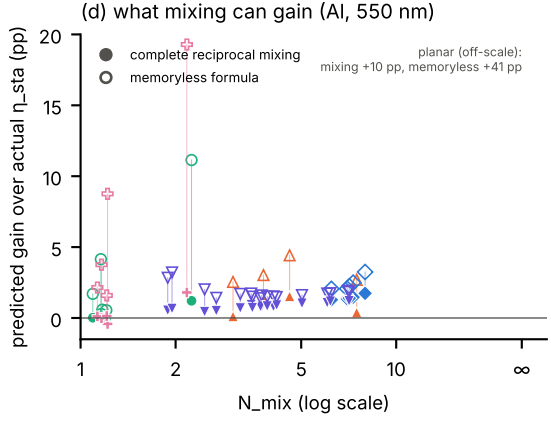

(a) structure map (η_sta, AI, 550 nm)
↑ planar: N_mix = ∞
0.60
0.70
0.75
0.78
0.80
0.82
planar
scattering layer
hemispherical MLA
freeform MLA
random MLA
wavy surface
angular mixing number N_mix (log scale)
escape probability p (uniform radiance)


(b) randomness vs angle retention
N_mix
scattering layer (g 0.8, τ* 1)
randomness parameter / its maximum
lenslet shape spread s (0–1)
position jitter / pitch (0–0.3)
wavy surface, rms slope (0–1.5)
height spread (0–0.6)


(c) late-pass escape
no dead-angle trapping
late-pass escape ratio ξ = p∞ / p̄
planar (N_mix = ∞, ξ = 0), drawn off-scale
N_mix (log scale)


(d) what mixing can gain (AI, 550 nm)
complete reciprocal mixing
memoryless formula
planar (off-scale): mixing +10 pp, memoryless +41 pp
predicted gain over actual η_sta (pp)
N_mix (log scale)